

Improving Customer Experience at McDonald's through Review Analysis and Clustering

Mini Project 3
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Problem

- How can McDonald's improve store experience and operational quality across different locations by analysing customer reviews?



Stakeholders

Store Managers

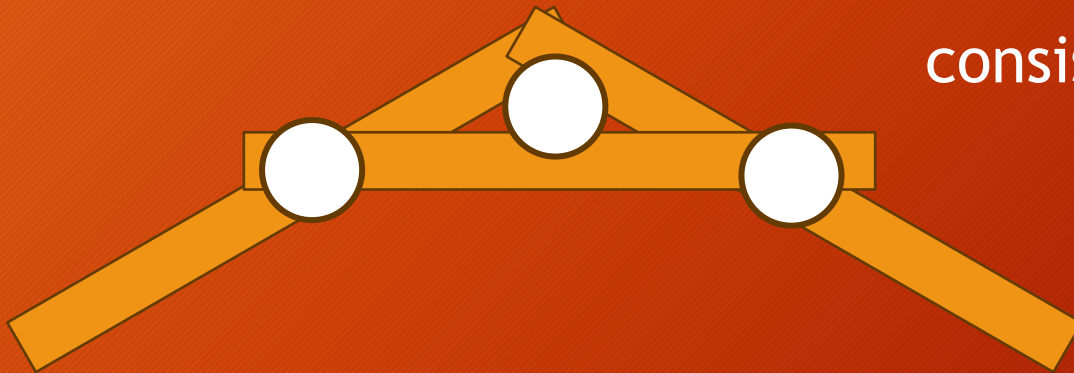
Reviews reflect how their stores are performing.

Marketing Teams

Positive or negative review trends affect the brand image.

Corporate Executives

Protect McDonald's reputation, increase profitability, and ensure a consistent brand experience.



Value

- **Operational efficiency:** Stores know what to fix (e.g. speed, cleanliness, staff issues).
- **Customer loyalty:** Happier customers are more likely to return.
- **Targeted training:** Train staff on the real issues being mentioned.
- **Understanding what customers like/dislike** = Craft better campaigns



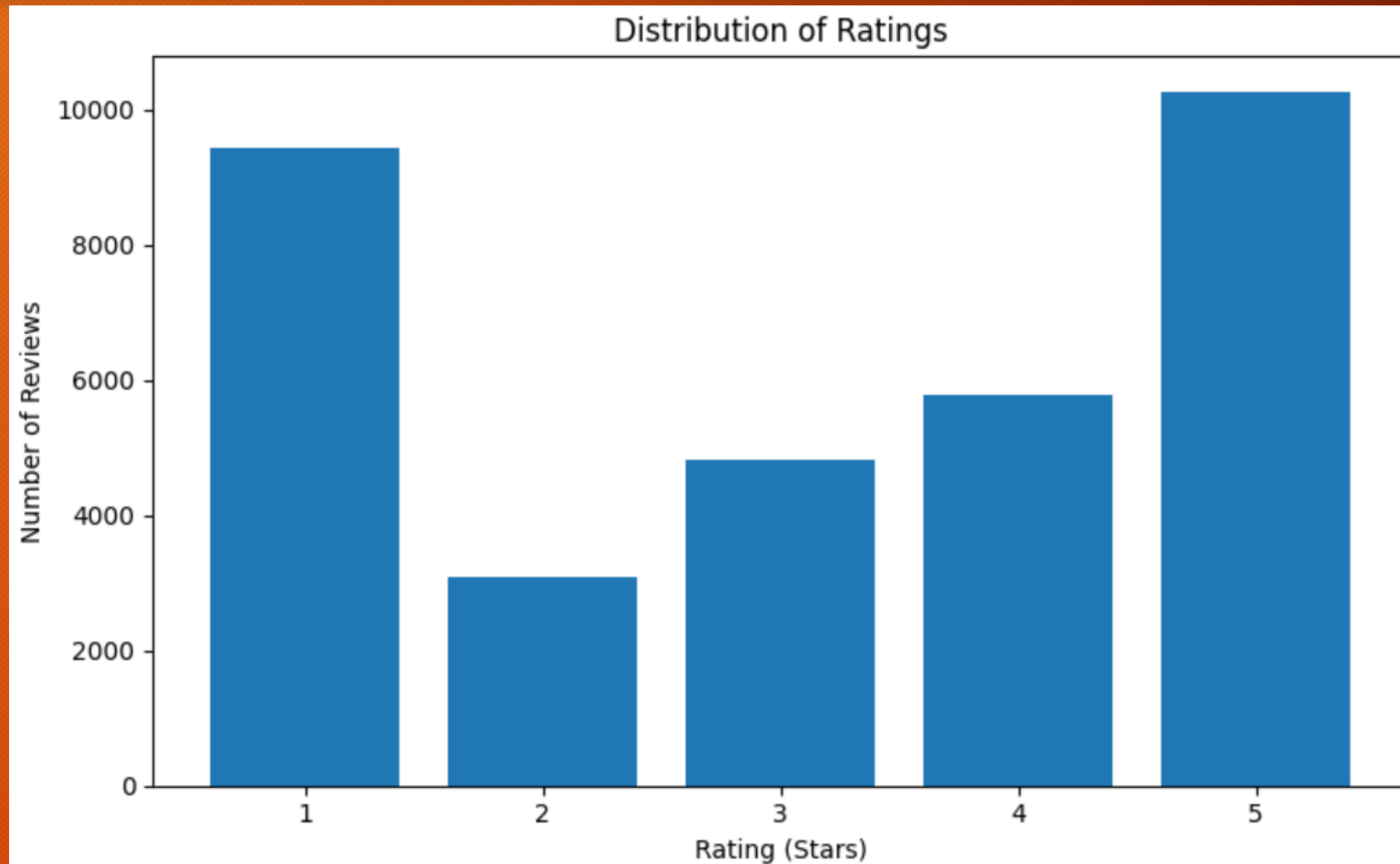
Data

- reviewer_id
 - store_name
 - category
 - **store_address**
 - latitude
 - longitude
 - rating_count
 - review_time
 - **review**
 - **rating**
- 33,396 anonymised reviews of McDonald's stores in the United States, scraped from Google reviews
 - <https://www.kaggle.com/datasets/nelgiriyeewithana/mcdonalds-store-reviews>

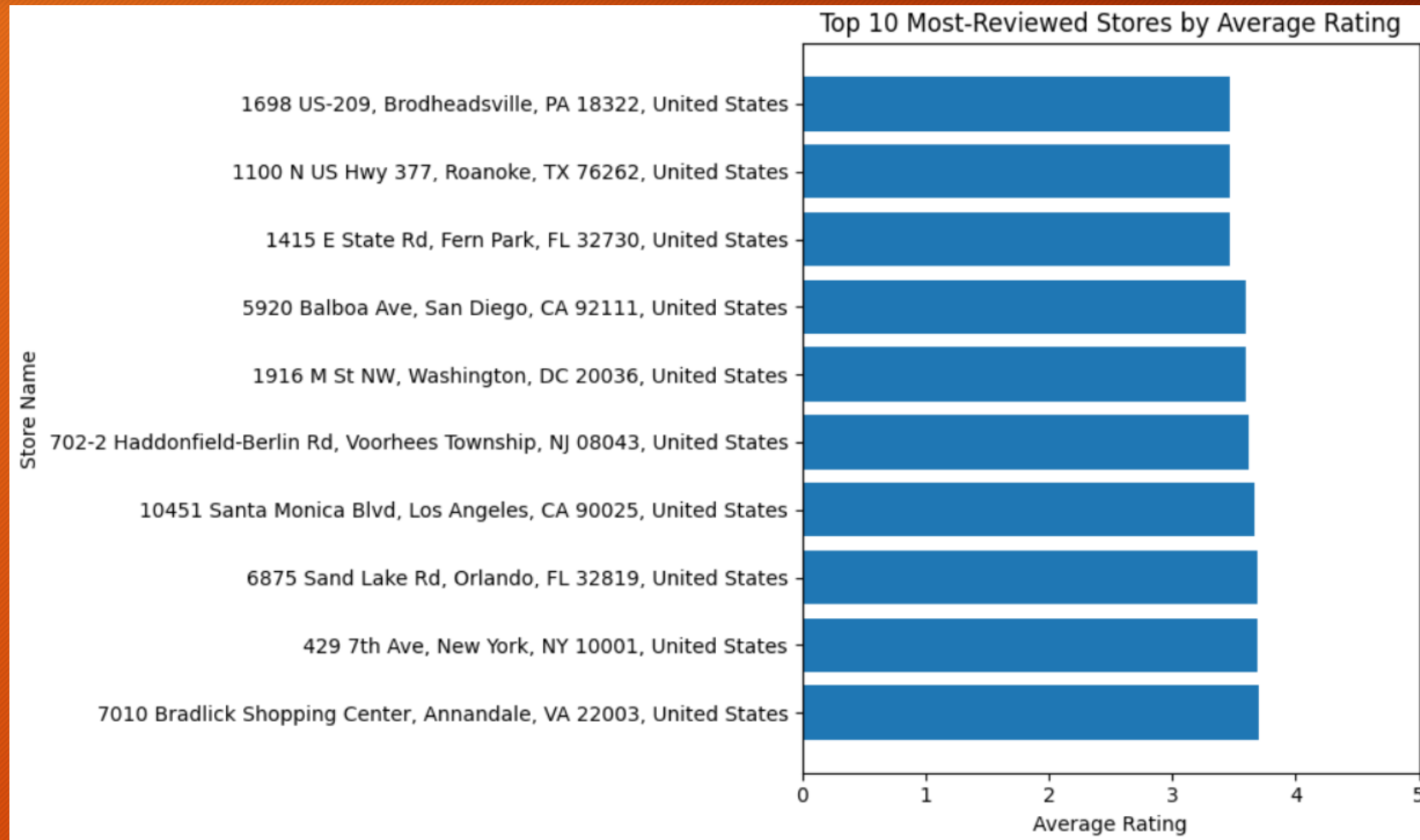


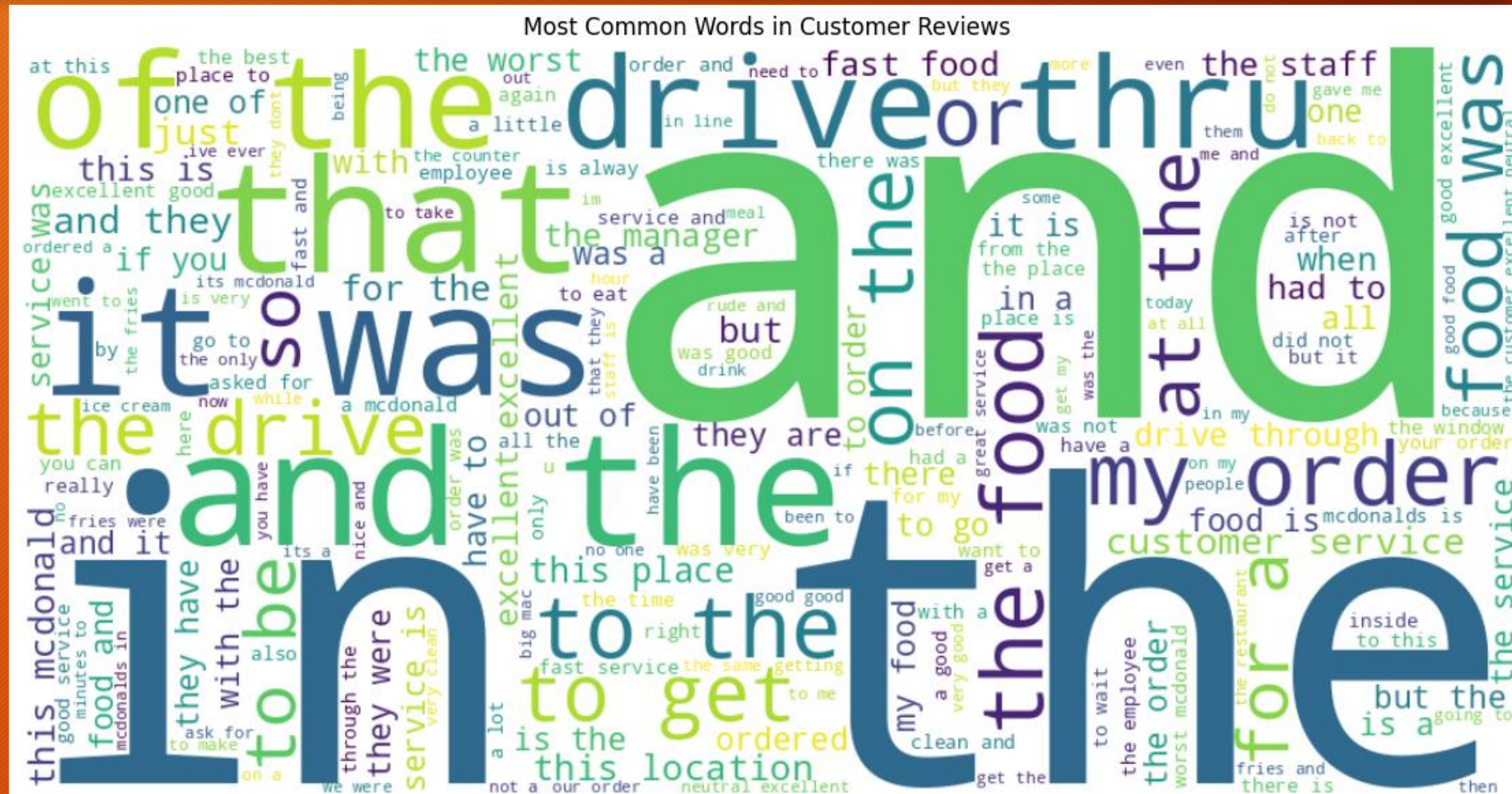
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Exploratory Data Analysis



Exploratory Data Analysis





Unsupervised learning model



Text Vectorisation (TF-IDF)

Term Frequency-Inverse Document Frequency converts words to vectors



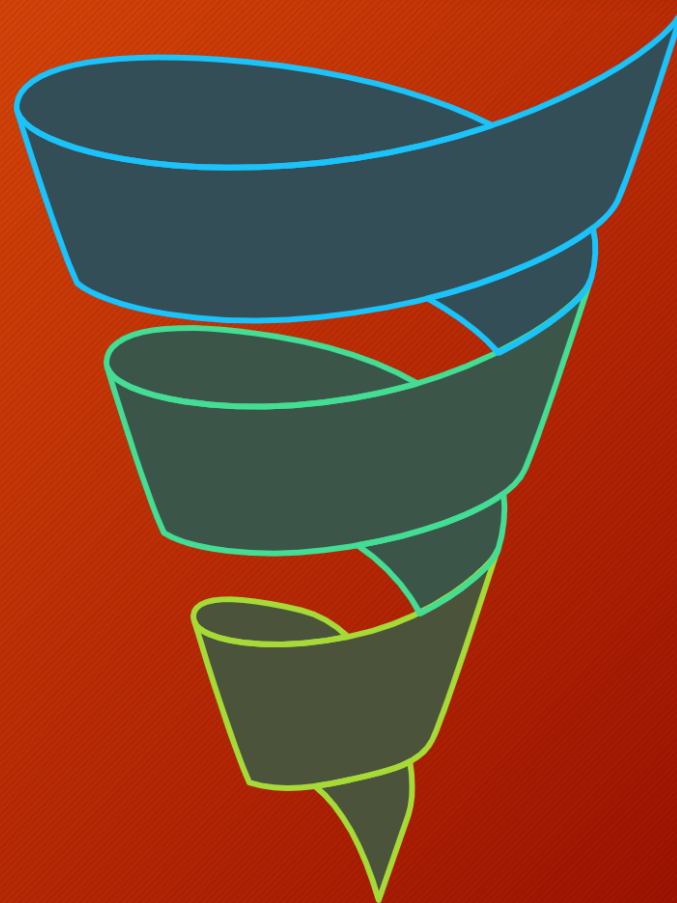
Dimensionality Reduction (PCA)

Principal Component Analysis to simplify the data while retaining key information



Clustering

Grouping similar data points together



Unsupervised learning model (K-Means)

Cluster	Top words	Theme
0	drive, fast, food, great, order, place	Speed, drive-thru service, and food quality
1	excellent, attention, customer	Highly positive reviews focused on excellent service
2	food, good, price, neutral	Comments about food quality and value
3	clean, fast, food, good, great, hot.	Positive emphasis on cleanliness and food freshness
4	clean, coffee, customer, food, fast, price	Service + beverage quality and customer care feedback

Topic Modelling (LDA)

Topic	Keywords	Interpretation
1	friendly, mcdonald, customer, quick, food, great, fast, service, excellent, good	Highly positive service experiences
2	time, open, service, mcdonalds, rude, long, order, drive, love, terrible	Mixed to negative experiences with drive-thru and customer service
3	worst, mcdonalds, ordered, minutes, got, just, like, fries, food, order	Strongly negative reviews.
4	fast, ok, mcdonalds, great, food, nice, staff, friendly, place, clean	Mildly positive or neutral experiences
5	food, service, hot, bad, best, slow, fresh, poor, mcdonalds, neutral	Mixed bag

Business recommendations

Store Managers

Cluster 0, 3, 4
Topic 1, 2



Customers value clean, fresh food, speed and accuracy in drive-thru, beverage quality.



Retrain staff on customer interaction and drive-thru efficiency



Investigate drive-thru delays, streamline ordering, improve staff response time.



Business recommendations

Marketing Teams

Cluster 1, 2, 3
Topic 1



Extremely positive reviews of service experience, consistency, speed.
Neutral/good value .



Feature these reviews and build campaigns around McDonald's consistency and speed



Target value-focused customers with promotions or loyalty perks.



Business recommendations

Corporate Executives

Cluster 0, 2, 4
Topic 2



Consistent complaints about speed and unfriendly service. Food quality inconsistencies.



Reinvest in technology (e.g. kitchen automation, app ordering) in slow stores.



Standardise operational procedures and training programs to reduce inconsistencies

